Reproducible Sparse-Concept and Calibration Diagnostics for Knowledge Tracing

A Protocol and Diagnostic Study

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Abstract

Knowledge Tracing (KT) models are commonly evaluated using aggregate predictive metrics such as AUC and accuracy. However, overall metrics may hide model behavior on sparse or limited cold-start Knowledge Components (KCs), where reliable probability estimates are crucial for educational decision support. This paper presents a reproducible diagnostic protocol for sparse-concept and calibration evaluation in KT. The protocol combines learner-based and temporal splits, train-only KC-frequency stratification, limited cold-start analysis, calibration metrics (ECE, Brier decomposition), reliability diagrams per stratum, and an eight-channel leakage and predictive sanity audit checklist. We instantiate the protocol on ASSISTments 2012, Junyi Academy, and XES3G5M with baselines IRT, DKT, and SimpleKT. Under our experimental conditions, on datasets with populated sparse strata, calibration error tends to increase from dense to sparse concepts, while aggregate AUC may obscure stratum-level reliability differences. We prepare scripts and report templates for release upon acceptance to support reproducible sparse-concept and calibration diagnostics for future KT studies.

Keywords: Knowledge Tracing; Educational Data Mining; Calibration; Sparse Concepts; Cold-start Knowledge Components; Reproducibility

1 Introduction

Knowledge Tracing (KT) estimates a learner's evolving mastery state from sequential interactions and predicts future correctness (???). Recent KT models, including memory-based, attention-based, and transformer-style variants, are typically compared using aggregate predictive metrics such as AUC and accuracy (???). These metrics are useful, but they may hide systematic weaknesses on low-frequency knowledge components (KCs), especially when most test interactions are concentrated on dense, frequently observed KCs.

This issue is not merely a reporting detail. In educational decision support, predicted probabilities can be used to decide whether a learner should receive remediation, practice another item, or progress to a new concept. A model that ranks answers well overall may still be poorly calibrated on sparse or limited cold-start KCs. In such cases, high predicted confidence can be misleading even when the model's aggregate AUC appears competitive. Calibration metrics such as Expected Calibration Error (ECE), Brier score, and reliability diagrams therefore provide a complementary diagnostic view (???).

This paper studies KT evaluation as a reproducibility and diagnostic problem. We do not propose a new KT architecture. Instead, we extend standard KT evaluation with train-only KC-frequency stratification, calibration diagnostics per stratum, limited cold-start KC analysis, and an eight-channel leakage and predictive sanity audit. This framing is intentionally conservative: the goal is to provide reusable evaluation infrastructure for future KT and structure-aware self-supervised learning studies, rather than to claim state-of-the-art performance.

1.1 Motivating Diagnostic Gap

Under standard benchmarking protocols, overall KT performance can be misleading because sparse KCs and limited cold-start KCs expose different predictive and calibration behavior. For example, a model may achieve competitive overall AUC on a learner-based split while showing substantially higher calibration error or unstable AUC estimates on sparse KC strata. This discrepancy is particularly critical in educational decision support, where predicted probabilities are used to make decisions regarding student remediation, progression, or mastery thresholds. If a model's probabilities are poorly calibrated, educational interventions may be triggered inappropriately, despite a deceptively high population-level discrimination score. This motivates a diagnostic protocol that reports performance and reliability at the KC-frequency level, coupled with a systematic leakage audit to ensure that evaluation pipelines do not inadvertently contaminate results.

In addressing this gap, this paper makes three contributions:

- Problem 1 (Hidden Stratum Behavior): Overall AUC/ACC values can mask noticeable degradation or metric instability on low-frequency concepts. Contribution 1: We formalize a train-only KC-frequency stratification protocol that prevents sparse-bucket leakage and enables sample-size-aware KT diagnostics across defined frequency buckets.

- Problem 2 (Unreliable Probability Estimates): Aggregate rankings do not reveal whether a model's predicted probabilities correspond to empirical correctness rates, which is crucial for educational decision-making. Contribution 2: We integrate calibration-oriented evaluation, including Expected Calibration Error (ECE), Brier decomposition (Uncertainty, Reliability, Resolution), and reliability diagrams, into bucket-level KT reporting.

- Problem 3 (Hidden Evaluation Pipeline Contamination): KT evaluation pipelines can suffer from subtle, undocumented data leakage across folds or splits. Contribution 3: We establish an eight-channel leakage and predictive sanity audit checklist and a reproducible workflow that exports prediction-level files, diagnostic reports, and paper-ready tables.

The remainder of this paper is organized as follows. Section 2 reviews KT evaluation, sparse-concept analysis, and calibration. Section 3 presents the diagnostic protocol. Section 4 reports the experimental design and diagnostic results. Section 5 discusses implications and limitations.

2 Related Work

2.1 Knowledge Tracing Models and Benchmarking

Knowledge Tracing (KT) formulates the task of modeling a learner's mastery over time to predict their probability of correctly answering future questions based on their interaction history (?). Classical approaches, such as Item Response Theory (IRT) (??) and Bayesian Knowledge Tracing (BKT) (?), model static ability and latent mastery states, respectively. ? introduced Deep Knowledge Tracing (DKT), shifting the paradigm toward recurrent neural networks. This was later followed by memory-augmented structures like DKVMN (?) and attention-based models such as AKT (?) and simpleKT (?). Recently, benchmarking toolkits such as pyKT (?) have significantly improved the standardization of data preprocessing and model evaluation. In this paper, we focus on diagnostics rather than proposing a new KT architecture; we utilize representative baselines to validate our protocols.

2.2 Graph-Augmented Knowledge Tracing

Recent advances in KT often incorporate auxiliary information using graph neural networks to model prerequisite, similarity, co-occurrence, or concept dependency relationships between knowledge components (KCs) (?). Models like GKT (?) and GIKT (?) construct KC-item or KC-KC dependency graphs, while contrastive/self-supervised approaches like CL4KT (?) leverage structurally augmented views. While these graph-augmented architectures effectively capture rich relations, they introduce significant complexity. Crucially, graph and self-supervised KT models require highly reliable evaluation protocols to prevent leakage and to report performance across varying KC-frequency strata. To maintain a focused evaluation on fundamental sequence modeling and calibration, our diagnostic protocol evaluates standard sequential KT models, establishing a rigorous foundation that can be extended to graph-based architectures in the future.

2.3 Sparse and Cold-start Knowledge Components

Educational datasets naturally exhibit long-tail distributions where a small fraction of dense KCs dominate interactions, leaving a vast majority of KCs with sparse training evidence (??). It is important to distinguish sparse and cold-start conditions in KT from typical cold-start scenarios in recommender systems. While recommender systems typically define cold-start at the user or item level (i.e., new users or items), KT sparse/cold-start issues relate to concepts (KCs) that have insufficient interaction history in the training data (??). Our work utilizes train-only KC-frequency stratification to separate strict cold-start concepts ($\mathbf{freq}_{\text{train}} = 0$) from very sparse, sparse, medium, and dense strata. This separation avoids conflating concepts with no training evidence with concepts that have limited but nonzero training evidence. This systematic stratification is designed to ensure that overall AUC metrics do not obscure model degradation on data-poor concepts.

Related KT models have also used the term sparse in a different sense. For example, sparseKT (?) introduces k-sparse attention to improve the robustness and generalization of attention-based KT models by selecting the most relevant historical interactions. This is conceptually different from our use of sparse KC: sparseKT sparsifies attention over interaction histories, whereas our protocol diagnoses low-frequency or zero-frequency knowledge components using train-only KC counts. Therefore, sparse attention and sparse concept evaluation should not be conflated.

Recent work has also addressed cold-start KT from complementary perspectives. For example, csKT (?) uses kernel bias and cone attention to improve KT under short student interaction sequences, where representations designed for longer histories may fail to capture early-stage KC relations. Other recent approaches use large language models to mitigate cold-start behavior, including CLST (?), which reformulates KT using natural-language descriptions of exercises and KCs, attribute-aware LLM-based KT (?), and option-weighted LLM-based KT (?). These directions are complementary to our protocol but target a different cold-start definition: they mainly address learner-level, sequence-level, system-level, or semantic cold-start, whereas our diagnostic protocol defines and audits cold-start at the concept/KC level through train-only frequency, with strict cold-start defined as $\mathsf{freq}_{\mathsf{train}}(c) = 0$. Because the present paper proposes a diagnostic protocol rather than a new KT architecture or attention mechanism, we do not compare directly against these model-level methods; instead, the proposed diagnostics can be applied to sparseKT-like and cold-start KT models in future work.

2.4 Calibration of Probabilistic Predictions

Model calibration is an essential property for KT models because their outputs—the predicted probability of correctness—can directly drive high-stakes educational interventions. In student modeling, Pelanek's evaluation framework is a central prior reference for assessing probabilistic student models using Brier score and calibration curves (?). This work establishes the importance of probability-level evaluation beyond binary accuracy. While deep neural networks can achieve high classification accuracy (e.g., high AUC), they may remain poorly calibrated, often producing overconfident predictions (?). In machine learning literature, calibration error is commonly quantified via Expected Calibration Error (ECE) (?). Furthermore, the Brier score (?) and its decomposition into uncertainty, reliability, and resolution (?) separate the inherent task difficulty from the model's predictive calibration. However, prior calibration-oriented student-model evaluation is typically not organized around train-only KC-frequency strata, strict concept-level cold-start groups, and sample-size-aware bucket diagnostics. Our protocol builds on this calibration tradition by reporting ECE, Brier decomposition, and reliability diagrams not only overall, but also across dense, medium, sparse, very sparse, and strict cold-start KC groups.

2.5 Reproducibility, Leakage Control, and Statistical Testing

The rapid proliferation of ML models has raised widespread concerns regarding reproducibility and data leakage (?), particularly in sequence modeling tasks like KT (?). Strict leakage control is necessary to prevent artificial inflation of performance. We identify and mitigate several leakage vectors, including split leakage, preprocessing leakage, KC mapping leakage, sparse-bucket leakage, calibration leakage, hyperparameter leakage, and cold-start leakage. For comparing competitive models, DeLong's test provides a standard approach for correlated ROC curves (?), and the Wilcoxon signed-rank test is appropriate for comparing repeated measurements (?). Statistical testing is used only where prediction-level outputs permit paired comparison; otherwise, we report descriptive diagnostics with sample-size-aware interpretation.

3 Diagnostic Protocol

3.1 Data Schema

Each dataset is normalized into a common interaction schema:

$$(u\_t, i\_t, c\_t, r\_t, \tau\_t), \tag{1}$$

where u_t is the learner, i_t is the item, c_t is the KC, $r\_t \in \{0,1\}$ is correctness, and τ_t is the timestamp. In this paper, multi-skill items are represented using a one-row-per-KC convention: each interaction involving multiple KCs is pre-expanded into one row per KC by the dataset source, so that each row in the processed data corresponds to exactly one (learner, item, KC, correctness, timestamp) tuple. Specifically, XES3G5M is loaded from the kc_level split provided by the dataset authors (?), which already stores one KC per row; ASSISTments 2012 is read from the standard release file in which each row carries a single skill_id; Junyi Academy items are inherently single-skill. No additional expansion is performed by our preprocess.py pipeline. KC-frequency stratification (Section 3.3) is computed on these pre-expanded rows using training-fold counts only.

3.2 Split Modes

We evaluate three split modes. The learner-based split assigns learners to train/validation/test partitions without learner overlap. The temporal split trains on earlier interactions and evaluates on later interactions. The limited cold-start concept split evaluates KCs with limited training-fold frequency. These modes answer different diagnostic questions and should not be conflated.

3.3 KC-frequency Stratification

For each training fold, we compute the training frequency of a KC c as

$$\text{freq\_train}(c) = \sum\_(u, i, c', r, \tau) \in D\_\text{train}\mathbb{I}[c' = c]. \tag{2}$$

KCs are assigned to five pre-registered groups:

$$\begin{aligned}
\text{Strict Cold-start:}\quad & \text{freq}_{\text{train}}(c) = 0, && (3)\\
\text{Very Sparse:}\quad & 0 < \text{freq}_{\text{train}}(c) < 20, && (4)\\
\text{Sparse:}\quad & 20 \le \text{freq}_{\text{train}}(c) < 100, && (5)\\
\text{Medium:}\quad & 100 \le \text{freq}_{\text{train}}(c) < 500, && (6)\\
\text{Dense:}\quad & \text{freq}_{\text{train}}(c) \ge 500. && (7)
\end{aligned}$$

The assignment must use only training-fold counts. Validation and test frequencies may be logged for auditing but must not determine the bucket. It is important to acknowledge a conceptual caveat regarding this definition: sparsity is not an absolute property of the underlying knowledge itself. Rather, it is partially an artifact of human authorship in the Q-Matrix construction. A concept might appear sparse not because the fundamental knowledge is rarely practiced, but because domain experts defined it at an overly granular level or tagged it inconsistently across items. Consequently, while our strata reliably isolate low-frequency training patterns, they reflect the operational sparsity of the dataset's specific KC mapping rather than an ontological truth about the learning domain.

3.4 Calibration Metrics

Given predictions p_i and labels y_i, ECE with $M = 15$ bins is

$$\text{ECE} = \sum\_m = 1^M \frac{|B\_m|}{N} \left|\text{acc}(B\_m) - \text{conf}(B\_m)\right|, \tag{8}$$

where $\text{acc}(B\_m)$ is empirical correctness and $\text{conf}(B\_m)$ is average predicted confidence in bin B_m.

The Brier score is

$$\text{Brier} = \frac{1}{N} \sum\_i = 1^N (p\_i - y\_i)^2. \tag{9}$$

We use the standard decomposition

$$\text{Brier} = \text{UNC} - \text{RES} + \text{REL}, \tag{10}$$

where the three components are defined as follows. Let $\bar{y}$ denote the global correctness rate, $M = 15$ the number of calibration bins, n_m the number of samples in bin m, acc_m the empirical accuracy in bin m, conf_m the average predicted confidence in bin m, and N the total number of samples.

$$\text{UNC} = \bar{y}(1 - \bar{y}). \tag{11}$$

$$\text{REL} = \frac{1}{N} \sum_{m=1}^{M} n_m \left(\text{conf}_m - \text{acc}_m\right)^2. \tag{12}$$

$$\text{RES} = \frac{1}{N} \sum_{m=1}^{M} n_m \left(\text{acc}_m - \bar{y}\right)^2. \tag{13}$$

UNC measures the intrinsic uncertainty of the prediction task and is independent of the model; REL measures the deviation between predicted probabilities and observed frequencies, where lower values indicate better reliability; RES measures how well the model separates events with different outcome rates from the global mean, where higher values indicate better resolution. Note that because this standard formulation relies on binning continuous probabilities into M discrete bins, the resulting decomposition is an empirical approximation and can be sensitive to binning artifacts. Calibration is reported both overall and per KC stratum.

AUC and calibration answer fundamentally different questions. AUC measures discrimination (i.e., whether a model ranks correct responses above incorrect ones), whereas calibration measures whether predicted probabilities correspond to empirical correctness rates. In educational decision support, this distinction is critical because probability thresholds are often used to trigger automated remediation, suggest additional practice, or license concept advancement. A model with high AUC may still exhibit substantial or potentially harmful miscalibration, leading to inappropriate instructional interventions due to biased confidence scores. Thus, calibration metrics are not auxiliary statistics but primary indicators of a model's operational reliability.

3.5 Eight-channel Leakage & Predictive Sanity Audit

Table 1 summarizes the leakage and predictive sanity audit. The audit is not a model component; it is an evaluation hygiene requirement.

Table 1: Leakage & Predictive Sanity Audit Checklist (L1–L8)

Channel	Risk / Description	Evidence	Status	Notes
L1	Split leakage (user overlap, temporal order)	split_audit.csv	PASS	No user overlap or temporal inversions detected
L2	Preprocessing leakage (transformations fit scope)	preprocess.py	PASS	No global normalization detected
L3	Q-matrix / KC mapping leakage	kc_map.json	PASS	Static mapping from dataset
L4	Sparse-bucket leakage	kc_strata.csv	PASS	Bucket assignment is strictly based on training frequency
L5	Calibration leakage (test-based tuning)	baseline_runner.py	PASS	No post-hoc tuning on test set
L6	Hyperparameter leakage (model selection)	baseline_runner.py	PASS	Validation-based selection only
L7	Cold-start leakage	split_constructor.py	PASS	Classification uses train_freq only
L8	Predictive sanity check	temporal_audit.py	PASS	Warm-cohort AUC vs static reference

3.6 Diagnostic Pipeline

Figure 1 illustrates the pipeline.

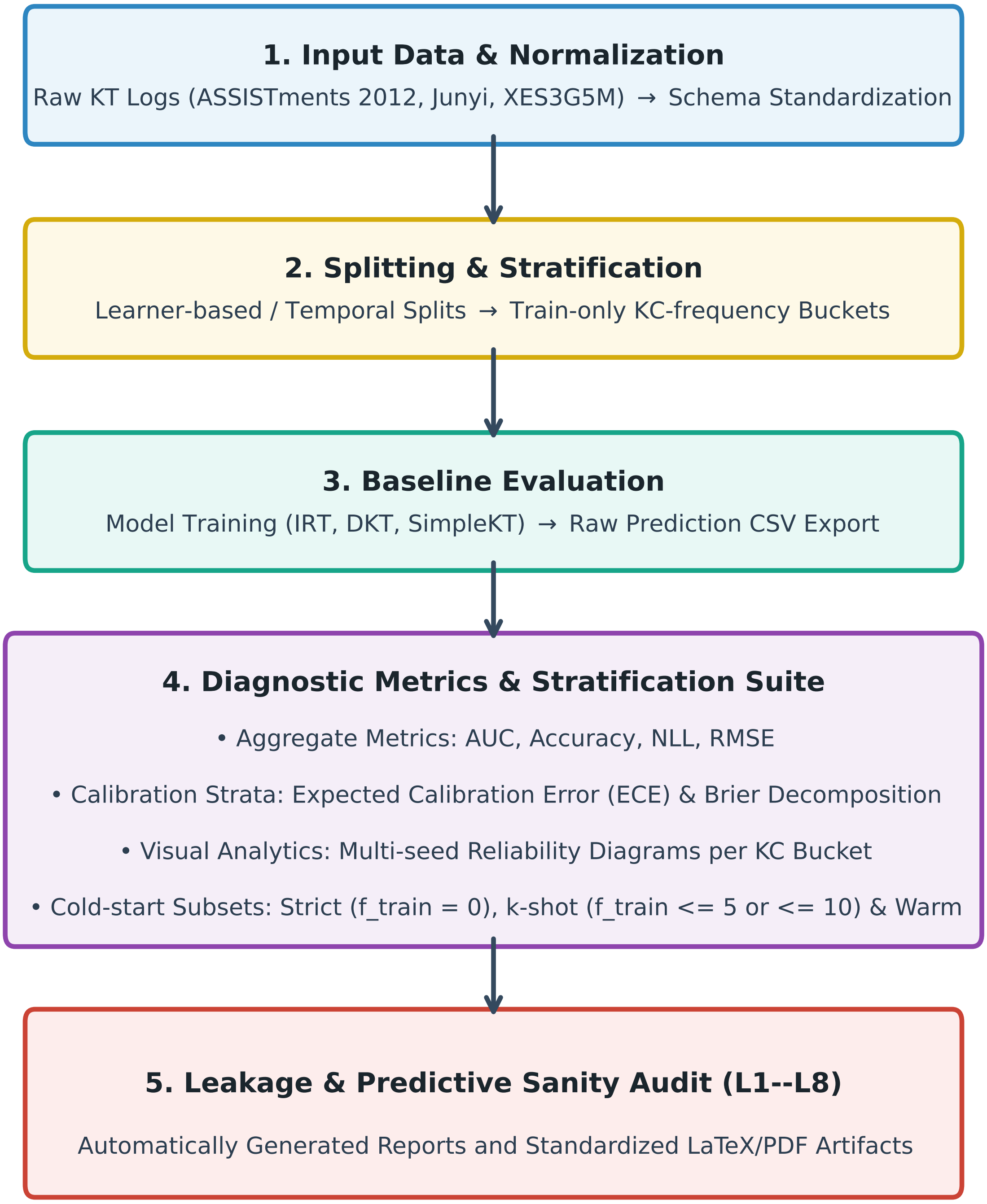


Figure 1: Overview of the reproducible sparse-concept and calibration diagnostic pipeline for Knowledge Tracing. (Note: In the figure, the term "k-shot" denotes "limited" cold-start KCs where $f\_\text{train} \leq 5$ or ≤ 10, as discussed in the text.)

3.7 Design Principles

To ensure the mathematical validity and operational reproducibility of the diagnostics, our protocol is governed by four core design principles:

- P1. Train-only Definitions: All knowledge component (KC) buckets and limited cold-start groups must be defined using training-fold information only. This prevents sparse-bucket leakage, wherein information about the testing split's distribution inadvertently influences the stratification boundaries or the preprocessing pipeline.

- P2. Prediction-level Export: Rather than computing downstream diagnostics in-memory during baseline training, our protocol mandates exporting raw prediction probabilities and ground-truth labels directly to disk. All metrics, including ECE, Brier score decomposition, and reliability diagrams, are computed post-hoc from these standardized artifacts. This decoupling helps keep diagnostic scoring consistent across baselines and reduces dependence on model-specific logging implementations.

- P3. Sample-size-aware Interpretation: Bucket-level metrics must never be reported in isolation. They must be accompanied by the exact number of KCs and interaction events contained within each stratum. This contextual reporting enables practitioners to differentiate between true performance degradation and metric instability due to small-sample constraints.

- P4. Leakage-audited Reporting: No diagnostic result is considered complete or valid unless accompanied by an explicit, verifiable leakage-control check. This is formalized through our eight-channel leakage and predictive sanity audit checklist (L1–L8), which enforces strict documentation of data splits, preprocessing scopes, and parameter configurations.

4 Experiments and Diagnostic Analysis

In this section, we instantiate our diagnostic evaluation protocol on three large-scale, benchmark Knowledge Tracing (KT) datasets. Our experimental investigation is structured around three core Research Questions (RQs):

- RQ1 (Predictive Strata Degradation): Do aggregate, population-level evaluation metrics (AUC, ACC) mask noticeable predictive degradation when models are evaluated on low-frequency (sparse and very sparse) Knowledge Components?

- RQ2 (Calibration Profiles by Frequency): How do model calibration profiles—measured via Expected Calibration Error (ECE) and Brier Score decomposition (Uncertainty, Reliability, Resolution)—change across different concept frequency strata?

- RQ3 (Limited Cold-start Concept Diagnostics): How do baseline KT models behave on KCs with zero or limited training-fold frequency under temporal splits?

4.1 Experimental Setup and Datasets

We evaluate our protocol on three diverse educational datasets: ASSISTments 2012 (assist2012), Junyi Academy (junyi), and XES3G5M (xes3g5m). The preprocessing and filtering pipeline (Section 3) ensures that only high-quality interaction records are retained. Table 2 presents the final statistics of each dataset cohort.

We use learner-based splits as the main predictive evaluation setting (reported in Tables 3 and 5, with detailed calibration results in Appendix C, Table 9) because they test generalization to unseen learners while preserving enough interactions for stable bucket-level analysis. Temporal splits are used as a complementary stress-test for future-oriented and limited cold-start diagnostics, reported in Table 6 and Appendix B. This explicit partitioning ensures that evaluation pipelines do not conflate learner-level and curriculum-level forecasting challenges.

We report mean $\pm$ standard deviation across five evaluation runs for all deep baselines on learner-based splits (folds corresponding to seeds 42, 2024, 2025, 2026, 2027). For temporal splits, after the label-alignment correction documented in Appendix F, we report deterministic metrics from the single corrected evaluation run (seed 42) used to regenerate Figure 3 and the temporal calibration tables. This single-run design is intentional: unlike learner-based cross-validation which randomly partitions students, temporal splitting relies on a single, deterministic chronological boundary (e.g., an 80/20 time cutoff) to simulate real-world future forecasting. Because the temporal partition is fixed, the reported temporal results focus on fixed-cutoff temporal stress testing rather than estimating variance across temporal cut-points or neural initializations. We do not claim that neural initialization variance is absent; extending the temporal diagnostics to multiple seeds or multiple temporal cut-points is left for future work. IRT is deterministic given a fold and its learner-based variations reflect fold differences, while its AUC in learner-based splits is static at 0.50 due to cold-start user constraints. Additionally, the #Events column denotes the number of prediction rows used for metric computation after model-specific prediction export and KC-strata matching. Following design principle P3, we annotate each bucket-level result with a reliability flag based on its sample size. Results from Insufficient buckets (N $<$ 100) are reported for completeness but should be interpreted as descriptive, not as evidence of model behavior.

Table 2: Dataset and Splitting Cohort Statistics

Dataset	Split Mode	#Learners	#Items	#KCs	#Interactions	#Train	#Valid	#Test	Avg. Seq. Len.
ASSISTments 2012	Learner-based	27,806	52,672	265	2,657,490	1,865,217	258,123	534,150	95.57
ASSISTments 2012	Temporal	27,806	52,672	265	2,657,490	1,860,242	265,749	531,499	95.57
Junyi Academy	Learner-based	71,014	25,784	1,326	16,215,567	11,373,990	1,572,555	3,269,022	228.34
Junyi Academy	Temporal	71,014	25,784	1,326	16,215,567	11,350,896	1,621,556	3,243,115	228.34
XES3G5M	Learner-based	18,066	7,653	866	7,953,709	5,572,247	792,317	1,589,145	440.26
XES3G5M	Temporal	18,066	7,653	866	7,953,709	5,567,596	795,370	1,590,743	440.26

Figure 2 illustrates heterogeneous KC-frequency distributions across datasets. Although dense KCs dominate the interaction volume, sparse and very sparse KCs remain important diagnostic strata because they expose low-frequency regions where aggregate metrics may be less informative.

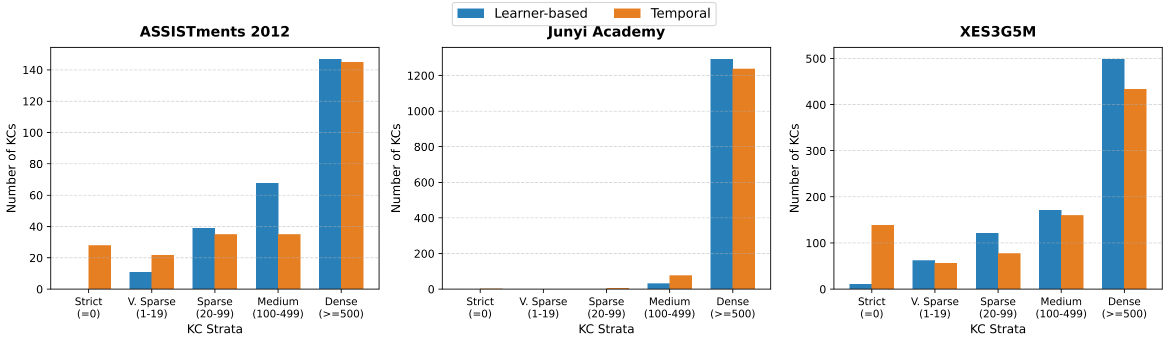


Figure 2: KC-frequency strata distribution across ASSISTments 2012, Junyi Academy, and XES3G5M. Strict Cold-start: $\text{train\_freq} = 0$; Very Sparse: $0 < \text{train\_freq} < 20$; Sparse: $20 \leq \text{train\_freq} < 100$; Medium: $100 \leq \text{train\_freq} < 500$; Dense: $\text{train\_freq} \geq 500$. Strict cold-start KCs are plotted separately from very sparse KCs to avoid conflating zero-frequency concepts with low-frequency concepts.

4.2 Baseline Implementations

We initially evaluated BKT but replaced it with IRT after observing numerical instability in EM-based pyBKT estimation; details are provided in Appendix E. We evaluate three representative classes of KT models:

1. Item Response Theory (IRT 1PL): A classical logistic baseline (specifically the one-parameter logistic Rasch model) that estimates a static user ability and concept difficulty. Because IRT does not model dynamic concept mastery transitions and relies on user identities, it cannot generalize to unseen users in learner-based splits (resulting in random ranking, AUC ≈ 0.50), but serves as a stable classical baseline under temporal splits.

2. Deep Knowledge Tracing (DKT): The pioneering recurrent neural network KT architecture mapping student sequential states via LSTM/RNN cells.

3. SimpleKT: A modern, self-attention-based model utilizing key-query similarities between concept embeddings to track mastery.

All deep learning models are trained with validation-based early stopping to prevent hyperparameter leakage, and we restrict our baselines strictly to official, un-augmented architectures to isolate diagnostic metrics.

4.3 RQ1: Strata Performance Analysis

To address RQ1, we first inspect the overall performance of all three baselines in Table 3. SimpleKT and DKT achieve high aggregate AUC scores under learner-based splits, with DKT slightly leading SimpleKT on ASSISTments 2012 under the learner-based split (0.6980 ± 0.0013 vs. 0.6840 ± 0.0025). Conversely, the classical IRT baseline performs at random (AUC $= 0.5000$) on learner-based splits. This is expected because IRT relies on student-specific latent traits that cannot be generalized to unseen learners in learner-based testing without initial observations. IRT's learner-based AUC remains at 0.50 because unseen learners do not have estimated ability parameters, causing the model to fall back to a constant global probability for all predictions. Consequently, its ACC simply reflects the majority-class base rate rather than any discriminative ranking

Table 3: Overall Performance under Learner-based Split

Dataset	Split	Model	AUC	ACC	NLL	RMSE
ASSISTments 2012	Learner-based	IRT	0.5000	0.6974 ± 0.0004	0.6131 ± 0.0003	0.4594 ± 0.0002
	Learner-based	DKT	0.6980 ± 0.0013	0.7183 ± 0.0013	0.5836 ± 0.0049	0.4399 ± 0.0011
	Learner-based	SimpleKT	0.6840 ± 0.0025	0.6999 ± 0.0034	0.7010 ± 0.0053	0.4600 ± 0.0022
Junyi Academy	Learner-based	IRT	0.5000	0.7050 ± 0.0017	0.6066 ± 0.0015	0.4561 ± 0.0008
	Learner-based	DKT	0.7317 ± 0.0012	0.7340 ± 0.0015	0.5446 ± 0.0026	0.4251 ± 0.0009
	Learner-based	SimpleKT	0.7228 ± 0.0035	0.7271 ± 0.0032	0.6100 ± 0.0216	0.4345 ± 0.0037
XES3G5M	Learner-based	IRT	0.5000	0.7957 ± 0.0028	0.5627 ± 0.0015	0.4316 ± 0.0008
	Learner-based	DKT	0.8170 ± 0.0028	0.8323 ± 0.0030	0.3940 ± 0.0085	0.3498 ± 0.0030
	Learner-based	SimpleKT	0.7559 ± 0.0013	0.8063 ± 0.0036	0.6215 ± 0.0141	0.3934 ± 0.0036

Note: IRT is used as the classical baseline across all datasets. Under learner-based splits, IRT's AUC remains at 0.5000 because unseen learners do not have estimated ability parameters, causing the model to fall back to a constant or base-rate-like prediction. Consequently, its ACC mainly reflects majority-class base-rate and thresholding behavior rather than discriminative ranking ability.

ability. Under temporal splits (Table 7 in Appendix B), IRT's AUC rises to 0.5913 on ASSISTments 2012 because the model evaluates seen students, while SimpleKT (0.6731) slightly outperforms DKT (0.6606), showing that model ordering can change across split modes.

Table 4: Pairwise DeLong Tests for Overall AUC (Learner-based Split)

Dataset	IRT vs DKT	IRT vs SimpleKT	DKT vs SimpleKT
ASSISTments 2012	$p < 0.001^*$	$p < 0.001^*$	$p < 0.001^*$
Junyi Academy	$p < 0.001^*$	$p < 0.001^*$	$p < 0.001^*$
XES3G5M	$p < 0.001^*$	$p < 0.001^*$	$p < 0.001^*$

Note: Bonferroni-corrected significance threshold: $\alpha = 0.05/9 = 0.0056$. * indicates significant after correction. IRT comparisons serve as the classical baseline reference; IRT's learner-based AUC remains at 0.5000 because unseen learners do not have estimated ability parameters, causing the model to fall back to a constant global probability for all predictions. Consequently, its ACC simply reflects the majority-class base rate rather than any discriminative ranking ability. DeLong tests are computed using paired prediction-level outputs aligned by test instances within each dataset.

Overall AUC differences between DKT and SimpleKT are statistically significant on all three datasets after Bonferroni correction, confirming that aggregate predictive quality differs between the two deep baselines. Classical baseline comparisons are reported for reference and highlight the generalization challenges on unseen cohorts.

However, when we break down the results by KC-frequency strata in Table 5, a far more nuanced picture emerges. Under our experimental conditions, deep KT models exhibit heterogeneous performance across strata. Specifically, while aggregate metrics suggest stable predictive power, bucket-level breakdowns in Table 5 show that in very sparse strata ($0 < f_{train} < 20$ interactions, excluding strict cold-start KCs), although DKT and SimpleKT show high AUC in the very sparse stratum on ASSISTments 2012 (e.g., 0.9045 ± 0.0667 for DKT), this estimate is based on an Insufficient sample size ($N < 100$) and should be interpreted descriptively rather than as stable evidence of model behavior. In contrast, medium and sparse strata do not show a simple monotonic AUC degradation pattern; instead, they exhibit heterogeneous behavior, including higher AUC in some strata but worse calibration, larger NLL/RMSE, or sample-size-sensitive instability. This supports our central point that aggregate AUC alone is insufficient for diagnosing KT reliability.

After test-fold filtering and KC-strata matching, the evaluated-KC totals were 260/265 for ASSISTments 2012, 1,326/1,326 for Junyi Academy, and 834/866 for XES3G5M. The evaluated-KC totals may be smaller than the dataset-level KC counts in Table 2 because only KCs appearing in the evaluated prediction exports after test-fold filtering and KC-strata matching are counted.

Table 5: Knowledge Tracing Performance Breakdown by Skill Strata (Learner-based)

Dataset	Model	Bucket	Rel.	#KCs	#Events	AUC	ACC	NLL	RMSE
ASSISTments 2012	IRT	dense	R	145	523,395	0.5000	0.6995 ± 0.0006	0.6113 ± 0.0005	0.4585 ± 0.0003
		medium	R	70	6,211	0.5000	0.5250 ± 0.0139	0.7560 ± 0.0115	0.5279 ± 0.0052
		sparse	L	37	403	0.5000	0.6253 ± 0.0503	0.6728 ± 0.0417	0.4889 ± 0.0200
		very_sparse	I	7	19	0.5000	0.4926 ± 0.0432	0.7829 ± 0.0358	0.5396 ± 0.0156
		strict_cold_start	I	1	4	-	0.0000	1.1915	0.6962
	DKT	dense	R	145	523,395	0.6969 ± 0.0012	0.7188 ± 0.0012	0.5803 ± 0.0037	0.4393 ± 0.0010
		medium	R	70	6,211	0.7455 ± 0.0107	0.6806 ± 0.0126	0.8201 ± 0.1010	0.4855 ± 0.0146
		sparse	L	37	403	0.7332 ± 0.0100	0.7051 ± 0.0229	1.1652 ± 0.1433	0.4980 ± 0.0106
		very_sparse	I	7	19	0.9045 ± 0.0667	0.7802 ± 0.0615	0.7088 ± 0.3389	0.4065 ± 0.0688
		strict_cold_start	I	1	4	-	0.5000 ± 0.7071	0.7013 ± 0.1371	0.5017 ± 0.0680
	SimpleKT	dense	R	145	523,395	0.6826 ± 0.0025	0.7001 ± 0.0035	0.6992 ± 0.0052	0.4597 ± 0.0023
		medium	R	70	6,211	0.7422 ± 0.0130	0.6809 ± 0.0112	0.8212 ± 0.0322	0.4833 ± 0.0082
		sparse	L	37	403	0.7197 ± 0.0183	0.6988 ± 0.0101	1.1214 ± 0.0774	0.4992 ± 0.0114
		very_sparse	I	7	19	0.8355 ± 0.0665	0.7678 ± 0.0751	0.9367 ± 0.2492	0.4609 ± 0.0677
		strict_cold_start	I	1	4	-	0.3750 ± 0.1768	0.7471 ± 0.0563	0.5252 ± 0.0256
Junyi Academy	IRT	dense	R	1,301	3,230,601	0.5000	0.7051 ± 0.0017	0.6065 ± 0.0015	0.4560 ± 0.0008
		medium	R	25	2,496	0.5000	0.5958 ± 0.0125	0.7006 ± 0.0108	0.5023 ± 0.0051
	DKT	dense	R	1,301	3,230,601	0.7319 ± 0.0012	0.7341 ± 0.0015	0.5441 ± 0.0026	0.4249 ± 0.0010
		medium	R	25	2,496	0.5954 ± 0.0358	0.5831 ± 0.0266	1.0982 ± 0.0827	0.5578 ± 0.0165
	SimpleKT	dense	R	1,301	3,235,313	0.7228 ± 0.0035	0.7272 ± 0.0032	0.6100 ± 0.0216	0.4344 ± 0.0037
		medium	R	25	2,689	0.6804 ± 0.0270	0.6449 ± 0.0291	0.6911 ± 0.0361	0.4810 ± 0.0142
XES3G5M	IRT	dense	R	497	1,269,390	0.5000	0.7959 ± 0.0028	0.5626 ± 0.0014	0.4315 ± 0.0008
		medium	R	174	12,977	0.5000	0.7834 ± 0.0050	0.5699 ± 0.0027	0.4356 ± 0.0015
		sparse	R	121	1,959	0.5000	0.7492 ± 0.0092	0.5898 ± 0.0053	0.4466 ± 0.0029
		very_sparse	L	40	106	0.5000	0.7189 ± 0.0291	0.6074 ± 0.0169	0.4560 ± 0.0090
		strict_cold_start	I	2	2	0.5000	0.5417 ± 0.3155	0.7111 ± 0.1838	0.5013 ± 0.0973
	DKT	dense	R	497	1,268,748	0.8168 ± 0.0025	0.8322 ± 0.0029	0.3923 ± 0.0073	0.3498 ± 0.0029
		medium	R	174	13,082	0.8587 ± 0.0089	0.8432 ± 0.0099	0.4996 ± 0.1066	0.3503 ± 0.0157
		sparse	R	121	2,002	0.8547 ± 0.0077	0.8264 ± 0.0138	0.7310 ± 0.1862	0.3786 ± 0.0164
		very_sparse	L	41	109	0.8172 ± 0.0553	0.7668 ± 0.0419	1.1867 ± 0.5323	0.4402 ± 0.0445
		strict_cold_start	I	2	2	0.7500 ± 0.3536	0.4583 ± 0.4167	0.6996 ± 0.0935	0.5019 ± 0.0453
	SimpleKT	dense	R	497	1,268,748	0.7548 ± 0.0013	0.8060 ± 0.0036	0.6214 ± 0.0139	0.3936 ± 0.0035
		medium	R	174	13,082	0.8344 ± 0.0049	0.8373 ± 0.0052	0.6195 ± 0.0337	0.3672 ± 0.0065
		sparse	R	121	2,002	0.8455 ± 0.0041	0.8167 ± 0.0043	0.6889 ± 0.0486	0.3852 ± 0.0063
		very_sparse	L	41	109	0.8188 ± 0.0341	0.7750 ± 0.0100	0.9877 ± 0.1510	0.4302 ± 0.0191
		strict_cold_start	I	2	2	0.5000 ± 0.7071	0.4167 ± 0.5000	0.9390 ± 0.4043	0.5858 ± 0.1509

Note: Reliability is assigned based on the number of test events: Reliable (R: $N \geq 1000$), Limited (L: $100 \leq N < 1000$), and Insufficient (I: $N < 100$). Results in Insufficient buckets are descriptive only. Bold results are not used in Insufficient buckets.

We note that the behavior observed in sparse and very sparse strata represents a combination of two distinct phenomena: (i) true performance degradation, where the model genuinely struggles to capture the mastery transitions of less-frequently observed concepts, and (ii) metric instability, where AUC or ACC values exhibit wide fluctuations due to the limited number of evaluation events. For instance, in very sparse strata under our experimental conditions, the number of test events can be as small as 3 to 10 events per seed. Consequently, bucket-level diagnostics should be interpreted jointly with sample size. Very sparse strata may yield unstable AUC estimates when the number of test events is very small. Therefore, in this protocol, the number of KCs and interaction events are not auxiliary information but required diagnostic context for a methodologically controlled, sample-size-aware evaluation.

An interesting counter-pattern emerges on XES3G5M. After separating strict cold-start KCs from the very-sparse stratum, DKT and SimpleKT still show higher AUC on sparse KCs than on dense KCs (DKT: 0.8547 vs. 0.8168; SimpleKT: 0.8455 vs. 0.7548). The very-sparse stratum remains competitive for both deep baselines (DKT: 0.8172; SimpleKT: 0.8188), but it should be interpreted cautiously because of its Limited reliability flag and small sample size. Strict cold-start rows are reported for completeness only and should not be used as stable evidence of model behavior. Several non-exclusive explanations for the counter-pattern are plausible: (i) XES3G5M's hierarchical KC tree structure may mean that sparse KCs are advanced concepts reached only by learners who have mastered prerequisites, creating a ceiling effect on accuracy; (ii) item-level features may carry more signal than KC-level features on XES3G5M, partially decoupling KC-frequency from predictive difficulty; and (iii) multi-skill expansion in XES3G5M may concentrate predictions on items whose primary skill is well trained even when the auxiliary skill is sparse. Disambiguating these explanations requires item-level error analysis beyond the scope of this protocol paper, but the pattern reinforces our point that bucket-level diagnostics reveal phenomena that overall AUC alone cannot.

4.4 RQ2: Calibration and Reliability Profiles

To answer RQ2, we analyze Expected Calibration Error (ECE) and Brier Score decomposition (Uncertainty UNC, Reliability REL, and Resolution RES) across frequency strata. On Junyi Academy under the learner-based split, the sparse and very sparse buckets are empty because all KCs have training-fold frequency at least 100. This absence is itself a diagnostic observation: Junyi does not exhibit learner-based sparse-KC concerns under the pre-registered thresholds.

Detailed calibration results by frequency stratum are reported in Appendix C, Table 9.

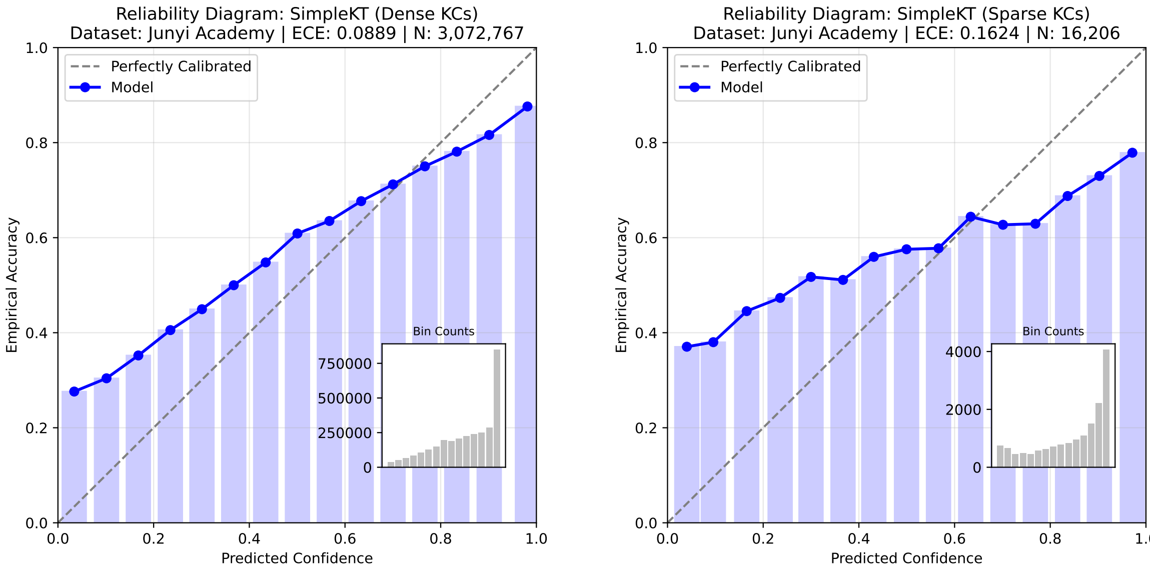
Notably, IRT exhibits very low ECE on several dense strata (e.g., 0.0033 on ASSISTments 2012 and 0.0021 on Junyi Academy) yet has zero Brier resolution (RES $= 0$) across strata and random discrimination (AUC $= 0.50$) on learner-based splits. This is the signature of a base-rate-like predictor: because IRT lacks ability estimates for unseen learners, it predicts close to the global correctness rate for most instances, yielding near-zero reliability error (REL ≈ 0, and hence Brier $\approx$ UNC) but no discriminative or resolving power. This concretely illustrates why calibration must be interpreted jointly with resolution and discrimination: a model can appear well-calibrated precisely by being uninformative.

Calibration gradient on ASSISTments 2012. Under our experimental conditions, ASSISTments 2012 provides the clearest example of the calibration gradient targeted by our protocol. For SimpleKT, ECE increases from 0.1131 in the dense stratum to 0.1578 in the medium stratum and 0.2254 in the sparse stratum. A similar monotonic degradation appears in the reliability component for DKT, where REL increases from 0.0053 (dense) to 0.0316 (medium) and 0.0623 (sparse). Since lower ECE and lower REL indicate better calibration, these results show that probability reliability can deteriorate as KC training evidence decreases, even when aggregate predictive metrics remain competitive. On the same dense stratum, IRT reports a much lower ECE (0.0033) than DKT (0.0602) and SimpleKT (0.1131); however, as discussed in the base-rate explanation above, this should not be interpreted as superior predictive reliability because IRT has zero resolution (RES $= 0$) and random learner-based discrimination (AUC $= 0.50$). This contrast illustrates that calibration diagnostics must be interpreted jointly with discrimination and resolution.

Our results suggest that, on datasets where sparse learner-based strata are present, calibration errors tend to increase as concept frequency decreases under our experimental conditions:

- On ASSISTments 2012, SimpleKT's ECE increases from 0.1131 ± 0.0076 in the dense stratum to 0.2254 ± 0.0204 in the sparse stratum, and is 0.2464 ± 0.0679 in the very sparse stratum.
- The reliability component (REL) of the Brier score also increases (lower is better), suggesting weaker calibration for sparse concepts under our experimental conditions.

Because temporal splits stress future-concept generalization, we use the Junyi temporal split in Figure 3 to visualize calibration behavior under a more challenging evaluation setting. Figure 3 compares SimpleKT reliability diagrams between dense and sparse KCs. The dense stratum is supported by substantially more test events and has lower ECE (0.0889), whereas the sparse stratum shows higher ECE (0.1624) while retaining meaningful discrimination (AUC = 0.6529). This comparison highlights why calibration and discrimination must be interpreted jointly at the stratum level.



(a) Reliability diagram of SimpleKT on Junyi temporal split for Dense KCs.

(b) Reliability diagram of SimpleKT on Junyi temporal split for Sparse KCs.

Figure 3: Comparison of reliability curves between Dense and Sparse KCs for SimpleKT on the Junyi temporal split after the label-alignment correction. The diagonal dashed line represents perfect calibration. Source calibration values are reported in Table 10. The sparse stratum has higher calibration error than the dense stratum while still retaining meaningful discrimination (AUC = 0.6529), illustrating that calibration degradation and discrimination should be interpreted jointly.

4.5 RQ3 (Limited Cold-start Concept Diagnostics): How do baseline KT models behave on KCs with zero or limited training-fold frequency under temporal splits?

Finally, we address RQ3 by evaluating models on limited cold-start concepts (strict, k5, k10, warm) under time-partitioned (temporal) splitting in Table 6.

Table 6: Cold-start Performance Metrics under Temporal Validation

Dataset	Model	Group	#KCs	#Events	AUC	ACC	ECE	Brier	REL	RES
ASSISTments 2012	IRT	strict	27	1,407	0.5000	0.5039	0.1987	0.2895	0.0395	0.0000
		k5	31	1,420	0.4937	0.5049	0.1998	0.2889	0.0403	0.0014
		k10	33	1,713	0.4070	0.4892	0.2143	0.2800	0.0523	0.0172
		warm	212	529,786	0.5919	0.6914	0.0207	0.2080	0.0008	0.0069
	DKT	strict	27	1,407	0.5055	0.5139	0.0598	0.2523	0.0046	0.0026
		k5	31	1,420	0.5093	0.5162	0.0598	0.2517	0.0051	0.0037
		k10	33	1,713	0.4910	0.5096	0.1225	0.2801	0.0529	0.0180
		warm	212	529,786	0.6606	0.6920	0.0844	0.2109	0.0117	0.0150
	SimpleKT	strict	27	1,407	0.4318	0.4634	0.1856	0.3005	0.0604	0.0101
		k5	31	1,420	0.4397	0.4676	0.1845	0.2988	0.0586	0.0099
		k10	33	1,713	0.4153	0.4530	0.2526	0.3456	0.1224	0.0220
		warm	212	529,786	0.6734	0.6804	0.1470	0.2273	0.0304	0.0172
Junyi Academy	IRT	strict	4	2,545	0.5000	0.5147	0.1934	0.2872	0.0374	0.0000
		k5	4	2,545	0.5000	0.5147	0.1934	0.2872	0.0374	0.0000
		k10	4	2,545	0.5000	0.5147	0.1934	0.2872	0.0374	0.0000
		warm	1,319	3,240,570	0.6526	0.7159	0.0220	0.1946	0.0007	0.0162
	DKT	strict	4	2,545	0.5903	0.5611	0.0494	0.2465	0.0051	0.0080
		k5	4	2,545	0.5903	0.5611	0.0494	0.2465	0.0051	0.0080
		k10	4	2,545	0.5903	0.5611	0.0494	0.2465	0.0051	0.0080
		warm	1,319	3,240,570	0.6949	0.7150	0.0746	0.1982	0.0092	0.0213
	SimpleKT	strict	4	2,545	0.5053	0.5501	0.0841	0.2536	0.0155	0.0106
		k5	4	2,545	0.5053	0.5501	0.0841	0.2536	0.0155	0.0106
		k10	4	2,545	0.5053	0.5501	0.0841	0.2536	0.0155	0.0106
		warm	1,319	3,240,570	0.7167	0.7199	0.0911	0.1945	0.0109	0.0264
XES3G5M	IRT	strict	117	233,214	0.5000	0.7875	0.2193	0.2154	0.0481	0.0000
		k5	125	258,943	0.5156	0.7691	0.2295	0.2177	0.0539	0.0009
		k10	126	259,425	0.5162	0.7689	0.2294	0.2177	0.0538	0.0009
		warm	424	1,331,318	0.6561	0.7822	0.0670	0.1590	0.0091	0.0074
	DKT	strict	117	233,214	0.4642	0.5010	0.2812	0.2549	0.0878	0.0007
		k5	125	258,943	0.4778	0.5253	0.2686	0.2472	0.0832	0.0011
		k10	126	259,425	0.4780	0.5257	0.2684	0.2471	0.0831	0.0011
		warm	424	1,331,318	0.6626	0.7456	0.1437	0.1875	0.0394	0.0092
	SimpleKT	strict	117	233,214	0.4773	0.5011	0.2973	0.2896	0.1223	0.0004
		k5	125	258,943	0.5015	0.5329	0.2774	0.2747	0.1109	0.0011
		k10	126	259,425	0.5017	0.5332	0.2773	0.2747	0.1108	0.0011
		warm	424	1,331,318	0.6615	0.7415	0.1860	0.2098	0.0607	0.0079

Note: For the Junyi Academy dataset, the strict, k5, and k10 cohorts coincide exactly because all 4 concepts in this category have zero training frequency. After applying the label-alignment correction consistently across datasets, warm-cohort temporal AUC recovers for deep KT baselines on ASSISTments 2012, Junyi Academy, and XES3G5M. Strict and limited-frequency cold-start groups remain more challenging and should be interpreted together with sample size and sanity-check evidence.

Under our experimental conditions, temporal splits expose a challenging generalization setting: both DKT and SimpleKT show limited generalization to KCs with zero or limited training-fold history, yielding near-random predictive performance (AUC $\approx$ 0.41 - 0.51) on strict and limited cold-start groups (e.g., DKT AUC is 0.5055 and SimpleKT AUC is 0.4318 on ASSISTments 2012 strict cold-start). This is expected because models lack parameter history to learn embeddings for unseen concepts. After applying the label-alignment correction documented in Appendix F consistently across datasets, deep KT baselines recover meaningful predictive signal on warm temporal cohorts. On Junyi Academy, DKT and SimpleKT reach warm-cohort AUCs of 0.6949 and 0.7167, respectively; on XES3G5M, they reach 0.6626 and 0.6615. On ASSISTments 2012, DKT reaches 0.6606 and SimpleKT reaches 0.6734. In contrast, strict and limited-frequency cold-start groups remain substantially more challenging, often yielding near-random or unstable AUC. This pattern indicates that the main temporal difficulty lies in generalizing to unseen or low-frequency KCs rather than in forecasting familiar KCs. Thus, temporal cold-start diagnostics help separate genuine concept-level cold-start failure from pipeline-induced random predictions.

4.6 Threshold Sensitivity Analysis

Threshold sensitivity analysis is provided as an optional robustness check in the released artifact. Because the present manuscript focuses on the pre-registered thresholds used throughout the main diagnostic tables, we do not use threshold sensitivity results as primary evidence. The main conclusions are therefore based on the pre-registered KC-frequency thresholds reported in Tables 5, 6, 9, and 10.

5 Discussion and Limitations

In this section, we discuss the broader implications of our diagnostic protocol for educational technology research and practice. We then define the scope boundaries of our work and address the primary limitations of the current study.

5.1 Implications

The proposed protocol encourages KT evaluation to move beyond aggregate AUC. Sparse-concept and calibration diagnostics can reveal whether a model is reliable where training evidence is limited. This is important for educational systems that use predicted probabilities for intervention or progression decisions.

5.2 Scope Boundaries

This paper is intentionally limited to evaluation diagnostics. It does not introduce a new KT model, a self-supervised learning objective, a graph neural network, a graph augmentation strategy, a learning path recommender, or a distillation method. Those directions require separate methodological papers and should be evaluated using the protocol introduced here. This paper focuses on downstream evaluation diagnostics for KT. A complementary line of work in our group addresses leakage control in concept-graph construction for graph-augmented KT. These two protocols are designed to be composable in future structure-aware KT studies.

5.3 Threats to Validity

We analyze the threats to the validity of our diagnostic protocol across standard dimensions plus temporal-alignment validity:

- Internal Validity: Risks to internal validity include potential preprocessing bugs, split leakage, sequence-label alignment errors (which motivated the explicit temporal alignment audit described below), and implementation-specific training differences across the IRT, DKT, and SimpleKT baselines. We mitigate these risks through a standardized interaction data schema, rigid validation-fold early stopping, and our automated leakage audit checklist.

- Temporal Alignment Validity: Sequence modeling in KT is notoriously prone to “silent bugs” such as off-by-one prediction-label misalignments. Our protocol actively defends against this via the warm-cohort predictive sanity audit (L8). During our temporal-split validation, this automated check successfully flagged an underlying library alignment discrepancy that otherwise would have systematically corrupted the evaluation. By isolating warm-cohort AUC before reporting cold-start metrics, the protocol provided a verifiable mechanism to detect and correct the artifact. This provides an empirical example supporting a core premise of our work: temporal cold-start results should not be interpreted in isolation; they should be read alongside explicit sanity audits to distinguish genuine concept-level cold-start difficulty from pipeline implementation errors.

- Construct Validity: Construct validity refers to whether our metrics accurately capture latent learner mastery and concept calibration. In KT, KC definitions depend heavily on dataset annotation quality, Q-matrix provenance, and how multi-skill items are processed. For example, expert-tagged KCs in ASSISTments 2012 and Junyi Academy may introduce varying granularities, which directly impact strata boundaries.

- External Validity: Our findings are based on three large-scale, public cohorts: ASSISTments 2012, Junyi Academy, and XES3G5M. The pronounced heterogeneity observed across these datasets—such as the clear calibration degradation gradient in ASSISTments 2012, the inverted calibration patterns in XES3G5M, and the entirely empty very-sparse stratum in Junyi Academy—highlights that sparse-concept behavior is heavily dataset-dependent. Rather than viewing this heterogeneity merely as a limitation to generalization, we argue it provides compelling empirical evidence for why a standardized, dataset-agnostic diagnostic protocol is necessary. Models cannot be assumed to behave consistently across different sparsity distributions, making localized validation using our protocol essential for transparent KT evaluation.

- Statistical Conclusion Validity: Statistical conclusion validity concerns the reliability of our metric estimates. As shown in Section 4, very sparse buckets contain very few active KCs and evaluation events (e.g., as few as 3 to 10 events). This sparsity inevitably introduces metric instability and high standard deviations, meaning strata differences must be interpreted in conjunction with explicit sample size counts.

6 Conclusion

This paper presents a reproducible diagnostic protocol for sparse-concept and calibration evaluation in Knowledge Tracing. The protocol combines train-only KC-frequency stratification, learner-based and temporal splits, limited cold-start concept analysis, ECE and Brier decomposition, reliability diagrams, and an eight-channel leakage and predictive sanity audit. The resulting artifact is intended to support fairer and more informative evaluation of KT models, especially future structure-aware and self-supervised approaches targeting sparse KCs. For instance, our experimental application of the protocol reveals that while aggregate population metrics may remain competitive, model calibration can degrade significantly as KC training evidence decreases—such as SimpleKT’s Expected Calibration Error (ECE) reliably degrading from 0.113 on dense concepts to 0.158 on medium concepts in ASSISTments 2012, and further increasing to 0.225 on sparse concepts (though interpreted with caution given its Limited reliability flag, $N = 403$). Furthermore, the observed heterogeneity in performance and calibration trends across different datasets underscores the necessity of a standardized diagnostic protocol. More broadly, our temporal-split audit demonstrates that the protocol's built-in leakage and sanity checks successfully catch subtle evaluation pipeline bugs, ensuring that researchers can distinguish genuine concept-level cold-start failure from alignment artifacts.

Artifact Availability & Reproducibility Checklist

The code, data preprocessing scripts, and reproducible evaluation pipelines are publicly available at https://github.com/trinhnkt/Sparse-Concept-and-Calibration. Consistent with the design principles of Section 3, the artifact includes the following standardized components:

- Dataset Preprocessing: Scripts that normalize raw data into the schema defined in Section 3.
- Split Construction: Reproducible partition logs and multi-seed split constructors.
- Prediction-level Exports: CSV files containing baseline predicted probabilities and ground-truth labels.
- KC-Strata Assignment: Code mapping training-fold frequencies to strata buckets.
- Calibration Verification: Scripts computing Expected Calibration Error (ECE).
- Brier Decomposition: Implementation of Uncertainty, Reliability, and Resolution equations.
- Reliability Diagrams: Plotting utilities for multi-strata calibration curves.
- Leakage & Predictive Sanity Audit: Automated logs verifying channels L1–L8.
- LaTeX Table Exporters: Python exporters generating book-tabs tables.
- One-Command Reproduction: Unified script ensuring machine-independent reproducibility.

A Threshold Sensitivity Protocol

Threshold sensitivity analysis is implemented in the released artifact as an optional robustness check. The present manuscript does not use threshold-sensitivity results as primary evidence because the main diagnostic conclusions are based on the pre-registered KC-frequency thresholds used in Tables 5, 6, 9, and 10. The artifact evaluates three alternative threshold configurations: (i) an aggressive setting with thresholds {10, 50, 250} training interactions, (ii) a conservative setting with thresholds {30, 150, 750} training interactions, and (iii) a dynamic quantile-based bucketing scheme where buckets are defined by equal-quantile KCs. These checks are intended to help future users evaluate whether their conclusions are sensitive to the chosen KC-frequency thresholds.

Table 7: Overall Performance under Future Validation (Temporal Splits)

Dataset	Split	Model	AUC	ACC	NLL	RMSE
ASSISTments 2012	Temporal	IRT	0.5913	0.6908	0.6049	0.4563
	Temporal	DKT	0.6606	0.6915	0.6362	0.4595
	Temporal	SimpleKT	0.6731	0.6796	0.7648	0.4771
Junyi Academy	Temporal	IRT	0.6525	0.7157	0.5738	0.4412
	Temporal	DKT	0.6949	0.7148	0.6181	0.4452
	Temporal	SimpleKT	0.7167	0.7198	0.6320	0.4411
XES3G5M	Temporal	IRT	0.6316	0.7801	0.5137	0.4106
	Temporal	DKT	0.6429	0.7097	0.6429	0.4441
	Temporal	SimpleKT	0.6433	0.7075	0.9014	0.4694

B Overall and Bucket-level Performance under Temporal Splits

In this section, we present the comprehensive overall performance (Table 7) and strata-specific performance breakdowns (Table 8) under the time-partitioned (temporal) split mode. As shown, deep learning baselines undergo noticeable predictive decay compared to their population-level learner-based counterparts, highlighting the increased difficulty of curriculum-level forecasting. Calibration degradation under temporal splits is reported separately in Appendix C, Table 10.

Table 8: Knowledge Tracing Performance Breakdown by Skill Strata (Temporal splits)

Dataset	Model	Bucket	Rel.	#KCs	#Events	AUC	ACC	NLL	RMSE
ASSISTments 2012	IRT	dense	R	145	521,661	0.5917	0.6929	0.6032	0.4554
		medium	R	29	4,558	0.5332	0.6038	0.6797	0.4924
		sparse	R	28	3,030	0.6486	0.5891	0.6789	0.4919
		very_sparse	L	16	843	0.7380	0.5302	0.6680	0.4875
		strict_cold_start	R	27	1,407	0.5000	0.5039	0.7794	0.5380
	DKT	dense	R	145	521,661	0.6619	0.6942	0.6256	0.4570
		medium	R	29	4,558	0.5637	0.5432	1.1556	0.5818
		sparse	R	28	3,030	0.6244	0.5743	1.3584	0.5859
		very_sparse	L	16	843	0.5971	0.4864	1.6419	0.6299
		strict_cold_start	R	27	1,407	0.5055	0.5139	0.6979	0.5023
	SimpleKT	dense	R	145	521,661	0.6745	0.6818	0.7556	0.4753
		medium	R	29	4,558	0.6084	0.5838	0.9636	0.5420
		sparse	R	28	3,030	0.6117	0.6023	1.6744	0.5800
		very_sparse	L	16	843	0.5617	0.5006	2.0213	0.6459
		strict_cold_start	R	27	1,407	0.4318	0.4634	0.8124	0.5482
Junyi Academy	IRT	dense	R	1,235	3,072,767	0.6483	0.7207	0.5692	0.4389
		medium	R	78	151,597	0.6487	0.6219	0.6582	0.4825
		sparse	R	6	16,206	0.6173	0.6703	0.6291	0.4679
		strict_cold_start	R	4	2,545	0.5000	0.5147	0.7752	0.5359
	DKT	dense	R	1,235	3,072,767	0.7073	0.7238	0.5784	0.4360
		medium	R	78	151,597	0.5562	0.5489	1.3060	0.5886
		sparse	R	6	16,206	0.5183	0.5930	1.6870	0.5857
		strict_cold_start	R	4	2,545	0.5903	0.5611	0.6862	0.4965
	SimpleKT	dense	R	1,235	3,072,767	0.7178	0.7259	0.6235	0.4369
		medium	R	78	151,597	0.6436	0.6046	0.7854	0.5122
		sparse	R	6	16,206	0.6529	0.6562	0.7917	0.4972
		strict_cold_start	R	4	2,545	0.5053	0.5501	0.7007	0.5036
XES3G5M	IRT	dense	R	340	1,191,744	0.6673	0.7993	0.4744	0.3893
		medium	R	72	129,494	0.5608	0.6227	0.6488	0.4778
		sparse	R	11	10,079	0.6903	0.8187	0.4720	0.3860
		very_sparse	R	10	26,212	0.6774	0.6037	0.6713	0.4879
		strict_cold_start	R	117	233,214	0.5000	0.7875	0.6236	0.4641
	DKT	dense	R	340	1,191,744	0.6679	0.7580	0.6029	0.4227
		medium	R	72	129,494	0.6214	0.6259	0.9269	0.5215
		sparse	R	11	10,079	0.6490	0.8228	0.5643	0.3785
		very_sparse	R	10	26,212	0.5529	0.7460	0.5518	0.4217
		strict_cold_start	R	117	233,214	0.4642	0.5010	0.7032	0.5049
	SimpleKT	dense	R	340	1,191,744	0.6666	0.7517	0.8803	0.4478
		medium	R	72	129,494	0.6136	0.6411	1.3648	0.5472
		sparse	R	11	10,079	0.6911	0.8286	0.9727	0.3862
		very_sparse	R	10	26,212	0.7308	0.8194	0.5170	0.3772
		strict_cold_start	R	117	233,214	0.4773	0.5011	0.7925	0.5381

C Detailed Calibration Results

Detailed calibration results by frequency stratum for the learner-based split are reported in Table 9. The corresponding temporal calibration breakdowns, computed after the label-alignment correction documented in Appendix F and used as the numerical basis for Figure 3, are reported in Table 10.

Table 9: Calibration Breakdown by Frequency Stratum

Dataset	Model	Bucket	Rel.	#Events	ECE	Brier	UNC	REL	RES
ASSISTments 2012	IRT	dense	R	523,395	0.0033 ± 0.0005	0.2102 ± 0.0002	0.2102	0.0000	0.0000
		medium	R	6,211	0.1711 ± 0.0139	0.2787 ± 0.0055	0.2492	0.0294	0.0000
		sparse	L	403	0.0708 ± 0.0503	0.2393 ± 0.0197	0.2323	0.0070	0.0000
		very_sparse	I	19	0.2036 ± 0.0432	0.2914 ± 0.0170	0.2485	0.0429	0.0000
		strict_cold_start	I	4	0.6962	0.4848	0.0000	0.4848	0.0000
	DKT	dense	R	523,395	0.0602 ± 0.0021	0.1930 ± 0.0009	0.2102	0.0053	0.0223
		medium	R	6,211	0.1658 ± 0.0270	0.2358 ± 0.0142	0.2492	0.0316	0.0442
		sparse	L	403	0.2320 ± 0.0109	0.2481 ± 0.0105	0.2323	0.0623	0.0457
		very_sparse	I	19	0.2079 ± 0.0424	0.1690 ± 0.0538	0.2485	0.0952	0.1733
		strict_cold_start	I	4	0.5014 ± 0.0678	0.2540 ± 0.0682	0.0000	0.2539	0.0000
	SimpleKT	dense	R	523,395	0.1131 ± 0.0076	0.2114 ± 0.0021	0.2102	0.0199	0.0187
		medium	R	6,211	0.1578 ± 0.0070	0.2336 ± 0.0079	0.2492	0.0289	0.0440
		sparse	L	403	0.2254 ± 0.0204	0.2493 ± 0.0115	0.2323	0.0587	0.0403
		very_sparse	I	19	0.2464 ± 0.0679	0.2161 ± 0.0643	0.2485	0.0830	0.1142
		strict_cold_start	I	4	0.5224 ± 0.0225	0.2761 ± 0.0269	0.0000	0.2758	0.0000
Junyi Academy	IRT	dense	R	3,230,601	0.0021 ± 0.0018	0.2080 ± 0.0007	0.2079	0.0000	0.0000
		medium	R	2,496	0.1071 ± 0.0127	0.2523 ± 0.0051	0.2407	0.0116	0.0000
	DKT	dense	R	3,230,601	0.0390 ± 0.0019	0.1806 ± 0.0008	0.2079	0.0022	0.0294
		medium	R	2,496	0.2588 ± 0.0219	0.3114 ± 0.0184	0.2407	0.0792	0.0079
	SimpleKT	dense	R	3,235,313	0.0804 ± 0.0068	0.1887 ± 0.0032	0.2079	0.0087	0.0277
		medium	R	2,689	0.1118 ± 0.0194	0.2315 ± 0.0136	0.2398	0.0163	0.0248
XES3G5M	IRT	dense	R	1,269,390	0.1542 ± 0.0031	0.1862 ± 0.0007	0.1624	0.0238	0.0000
		medium	R	12,977	0.1416 ± 0.0053	0.1898 ± 0.0013	0.1697	0.0201	0.0000
		sparse	R	1,959	0.1074 ± 0.0094	0.1995 ± 0.0026	0.1879	0.0116	0.0000
		very_sparse	L	106	0.0772 ± 0.0291	0.2080 ± 0.0082	0.2014	0.0066	0.0000
		strict_cold_start	I	2	0.2794 ± 0.0950	0.2584 ± 0.0894	0.1736	0.0848	0.0000
	DKT	dense	R	1,268,748	0.0389 ± 0.0023	0.1223 ± 0.0020	0.1617	0.0027	0.0415
		medium	R	13,082	0.0906 ± 0.0195	0.1229 ± 0.0111	0.1684	0.0133	0.0576
		sparse	R	2,002	0.1205 ± 0.0195	0.1435 ± 0.0123	0.1878	0.0197	0.0628
		very_sparse	L	109	0.2009 ± 0.0455	0.1954 ± 0.0396	0.2042	0.0631	0.0698
		strict_cold_start	I	2	0.4259 ± 0.1454	0.2534 ± 0.0462	0.1181	0.2094	0.0764
	SimpleKT	dense	R	1,268,748	0.1146 ± 0.0010	0.1550 ± 0.0028	0.1617	0.0179	0.0238
		medium	R	13,082	0.1117 ± 0.0060	0.1349 ± 0.0048	0.1684	0.0168	0.0491
		sparse	R	2,002	0.1244 ± 0.0074	0.1484 ± 0.0049	0.1878	0.0212	0.0595
		very_sparse	L	109	0.1951 ± 0.0186	0.1854 ± 0.0163	0.2042	0.0585	0.0766
		strict_cold_start	I	2	0.5836 ± 0.1525	0.3603 ± 0.1819	0.1181	0.3603	0.1181

Note: IRT shows low ECE in several learner-based cohorts, but RES = 0 across strata indicates base-rate-like behavior with no resolving power. Thus, IRT calibration should be interpreted jointly with AUC and Brier resolution.

Table 10: Calibration Breakdown by Frequency Stratum under Temporal Splits. Values are computed from prediction-level outputs after the label-alignment correction documented in Appendix F. Very Sparse excludes Strict Cold-start KCs; Strict Cold-start is reported separately where applicable. This table provides the source calibration values used for Figure 3.

Dataset	Model	Bucket	Rel.	#Events	ECE	Brier	UNC	REL	RES
ASSISTments 2012	IRT	dense	R	521,661	0.0194	0.2074	0.2138	0.0007	0.0069
		medium	R	4,558	0.0853	0.2425	0.2347	0.0101	0.0027
		sparse	R	3,030	0.1374	0.2419	0.2340	0.0261	0.0179
		very_sparse	L	843	0.2492	0.2376	0.2042	0.0674	0.0331
		strict_cold_start	R	1,407	0.1987	0.2895	0.2500	0.0395	0.0000
	DKT	dense	R	521,661	0.0812	0.2088	0.2138	0.0104	0.0154
		medium	R	4,558	0.2796	0.3385	0.2347	0.1068	0.0036
		sparse	R	3,030	0.3149	0.3433	0.2340	0.1228	0.0127
		very_sparse	L	843	0.3553	0.3967	0.2042	0.2100	0.0162
		strict_cold_start	R	1,407	0.0598	0.2523	0.2500	0.0046	0.0026
	SimpleKT	dense	R	521,661	0.1454	0.2260	0.2138	0.0297	0.0174
		medium	R	4,558	0.2277	0.2938	0.2347	0.0677	0.0089
		sparse	R	3,030	0.3069	0.3363	0.2340	0.1120	0.0093
		very_sparse	L	843	0.3879	0.4172	0.2042	0.2322	0.0189
		strict_cold_start	R	1,407	0.1856	0.3005	0.2500	0.0604	0.0101
Junyi Academy	IRT	dense	R	3,072,767	0.0233	0.1926	0.2070	0.0008	0.0150
		medium	R	151,597	0.0625	0.2328	0.2479	0.0051	0.0201
		sparse	R	16,206	0.0398	0.2189	0.2306	0.0020	0.0136
		strict_cold_start	R	2,545	0.1934	0.2872	0.2498	0.0374	0.0000
	DKT	dense	R	3,072,767	0.0629	0.1901	0.2070	0.0062	0.0230
		medium	R	151,597	0.2921	0.3465	0.2479	0.1007	0.0023
		sparse	R	16,206	0.3131	0.3430	0.2306	0.1130	0.0010
		strict_cold_start	R	2,545	0.0494	0.2465	0.2498	0.0051	0.0080
	SimpleKT	dense	R	3,072,767	0.0889	0.1909	0.2070	0.0102	0.0262
		medium	R	151,597	0.1552	0.2623	0.2479	0.0299	0.0154
		sparse	R	16,206	0.1624	0.2472	0.2306	0.0324	0.0158
		strict_cold_start	R	2,545	0.0841	0.2536	0.2498	0.0155	0.0106
XES3G5M	IRT	dense	R	1,191,744	0.0547	0.1516	0.1563	0.0046	0.0092
		medium	R	129,494	0.1759	0.2283	0.1691	0.0616	0.0026
		sparse	R	10,079	0.1823	0.1490	0.1202	0.0368	0.0080
		very_sparse	R	26,212	0.3189	0.2380	0.1404	0.1061	0.0085
		strict_cold_start	R	233,214	0.2193	0.2154	0.1673	0.0481	0.0000
	DKT	dense	R	1,191,744	0.1315	0.1787	0.1563	0.0324	0.0099
		medium	R	129,494	0.2573	0.2720	0.1691	0.1097	0.0065
		sparse	R	10,079	0.1234	0.1433	0.1202	0.0307	0.0076
		very_sparse	R	26,212	0.1549	0.1779	0.1404	0.0420	0.0047
		strict_cold_start	R	233,214	0.2812	0.2549	0.1673	0.0878	0.0007
	SimpleKT	dense	R	1,191,744	0.1756	0.2006	0.1563	0.0532	0.0085
		medium	R	129,494	0.2855	0.2994	0.1691	0.1349	0.0044
		sparse	R	10,079	0.1405	0.1491	0.1202	0.0391	0.0099
		very_sparse	R	26,212	0.1002	0.1423	0.1404	0.0137	0.0109
		strict_cold_start	R	233,214	0.2973	0.2896	0.1673	0.1223	0.0004

Note: These values serve as the numerical basis for Figure 3. IRT calibration should be interpreted jointly with discrimination and Brier resolution. Under learner-based splits, IRT behaves like a base-rate predictor for unseen learners; under temporal splits, it can recover non-zero resolution because learners are observed during training.

D Diagnostic Interpretation Guide

To assist researchers and practitioners in applying our diagnostic protocol, we provide Table 11 which maps common empirical diagnostic patterns observed during evaluation to their theoretical and operational educational interpretation.

Table 11: Diagnostic Interpretation Guide for Sparse-Concept and Calibration Analysis

Diagnostic Pattern	Interpretation / Recommendation
1. High overall AUC but high sparse-KC ECE	The model ranks responses well overall (good discrimination) but its predicted probabilities may be potentially unreliable (poor calibration) on low-frequency concepts. Educational interventions triggered by probability thresholds should be used with caution on these KCs.
2. Dense AUC remains stable but very-sparse AUC is unstable or has high variance (e.g., high standard deviations)	Check the number of KCs and interaction events in the stratum. Volatility in metrics can be driven by small sample-size constraints (metric instability) rather than intrinsic model shifts.
3. Brier reliability (REL) component increases in sparse buckets	Probabilistic calibration degrades under limited training evidence. Under our experimental conditions, deep sequence models tend to generate overconfident or under-confident probability predictions as concepts become scarcer.
4. Temporal split AUC converges near random guess (≈ 0.50), especially on strict cold-start groups	Future-oriented concept generalization can be challenging, especially under strict temporal cold-start constraints. However, if near-random AUC occurs on warm or dense cohorts where sufficient training evidence exists, the L8 predictive sanity check should be triggered to audit possible prediction–label alignment, sequence cut-off, export, or metric-computation errors before interpreting the result as genuine model failure.
5. strict, k5, and k10 cold-start groups show near-identical event counts or metrics	The dataset under study has very few genuinely cold-start concepts under the chosen split, or the KCs are densely linked, meaning the cold-start setting is not highly active for this cohort.
6. ECE and Brier score model rankings differ from AUC discrimination rankings	Discrimination and calibration capture orthogonal model characteristics. They answer different diagnostic questions and should be reported separately. Models optimized purely for pairwise ranking (AUC) may not produce reliable soft posterior masteries.

E Bayesian Knowledge Tracing Numerical Instability and Transition

During our initial protocol implementation, we instantiated Bayesian Knowledge Tracing (BKT) as the classical baseline utilizing the popular pyBKT library (v1.4.1). However, on all three large-scale datasets, BKT suffered from severe numerical instability during Expectation-Maximization (EM) parameter estimation. Specifically, for sparse and very sparse concept strata, the scarcity of interaction logs triggered a divide-by-zero error in the M-step update equation for the prior parameter, resulting in NaN value assignments. Consequently, the model's predictions degenerated into deterministic binary outputs ($p_{pred} \in \{0.0, \mathrm{NaN}\}$), yielding a flat 0.50 AUC and extreme Negative Log-Likelihood (NLL) values ranging between 23.7 and 28.0. To maintain scientific integrity and standard baseline comparison, we transitioned to Item Response Theory (IRT 1PL / Rasch Model) as the primary classical baseline. The raw prediction CSV files for the failed BKT runs are preserved within the codebase predictions directory for transparency.

F Prediction–Label Alignment Audit for Temporal Splits

During the temporal-split audit, we identified a prediction–label alignment issue in the temporal evaluation pipeline. The issue occurred when exported model predictions were matched back to temporal test instances after sequence construction and cohort filtering. In the affected code path, prediction rows and correctness labels could be aligned by transient row order rather than by a stable instance identifier, which made some exported p_{pred} values correspond to incorrect y_{true} entries.

The main symptom was not limited to strict cold-start KCs. Deep KT baselines produced near-random AUC even on warm temporal cohorts, where KCs had sufficient training-fold evidence. This behavior was a red flag because a static IRT baseline still achieved meaningful AUC on the same warm cohorts, whereas DKT and SimpleKT collapsed to approximately random ranking.

We detected the issue using two sanity checks. First, we compared warm-cohort AUC against the IRT reference baseline; near-random deep KT AUC on warm KCs was treated as an alignment warning rather than as evidence of genuine temporal generalization failure. Second, we computed prediction–label correlation on warm-cohort slices and verified whether $\mathbb{E}[p_{pred} \mid y=1]$ exceeded $\mathbb{E}[p_{pred} \mid y=0]$. The correction preserves a stable test-instance identity through temporal splitting, sequence construction, prediction export, cohort filtering, and metric computation. Metrics are computed only after pairing predictions and labels by this stable identity rather than by transient row position.

After applying the correction consistently across datasets, warm-cohort AUC for the affected deep KT baselines recovered above the sanity threshold, and prediction–label correlations became positive. These checks were then used before re-running the temporal diagnostics reported in the paper. This audit reinforces the central message of the protocol: temporal cold-start results should be accompanied by warm-cohort and prediction–label sanity checks to distinguish genuine concept-level cold-start difficulty from alignment artifacts.

Declaration of Generative AI Software Tools in the Writing Process

During the preparation of this work, the authors used ChatGPT and Google Antigravity to support language polishing, LaTeX formatting, consistency checking, and preparation of reproducibility prompts in the Abstract, Introduction, Methods, Results, Discussion, and Appendix sections. These tools were not used to generate, fabricate, or alter experimental results. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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